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EDUCATION

Northeastern University

M.S. Computer Science — AI and Software Engineering

Coursework: Algorithms, Building Scalable Distributed Systems, Database Management Systems, Programming Design Paradigms

Seattle, WA

Sep 2025 – May 2027

Siddaganga Institute of Technology

B.Tech, Artificial Intelligence and Data Science

Tumkur, India

Dec 2021 – Jun 2025

PROJECTS

Helmsman — LLM Inference Gateway

2025

Go, Redis, Prometheus, Grafana, Kubernetes | github.com/sohanreddyk/helmsman

- Built an OpenAI-compatible LLM inference gateway in Go serving **594 RPS warm vs. 34 RPS cold — a 17× throughput gain** — via semantic caching (Ollama embeddings, cosine ≥ 0.95) at 0.2 ms p50 cache-hit latency, with async cache writes kept off the response path.
- Engineered lock-free round-robin load balancing (atomic.Uint64), per-key Redis token-bucket rate limiting (atomic Lua), and **per-backend circuit breakers** (Closed/Open/Half-Open, 3 failures / 30 s cooldown); deployed on Kubernetes with HPA.
- Designed a bounded-concurrency semaphore returning 503 on saturation with exponential-backoff retry across healthy backends; ran health probes outside the write lock to prevent latency spikes.

Lumen — Distributed AI Evaluation Platform

2025

Python, FastAPI, Celery, PostgreSQL, Redis, React, Kubernetes | github.com/sohanreddyk/lumen

- Built a distributed LLM evaluation platform scoring model outputs across **5 pluggable evaluators** (exact match, keyword, regex, embedding similarity, LLM-as-judge); Celery workers (Redis-backed, acks_late) pull runs from a queue and persist to PostgreSQL.
- Implemented cross-run **regression detection** via a Mann-Whitney U test (p-value + rank-biserial effect size, gated on $n \geq 5$), auto-alerting when **pass rate drops below 70%** — surfaced on a live-polling React dashboard.
- Added a zero-latency live traffic mirror (FastAPI BackgroundTasks) recording prompt/response/latency async to PostgreSQL; instrumented workers with Prometheus counters and histograms, deployed on Kubernetes with HPA (2 → 6 replicas).

Distributed Chat Server

2025

Java, WebSocket, RabbitMQ, AWS EC2

- Built a multithreaded Java WebSocket chat server on AWS EC2 with a RabbitMQ topic exchange for message routing and channel pooling to cut broker connection overhead.
- Optimized thread-pool sizing and connection reuse to sustain **~131K messages/sec across 4 load-balanced servers** (RabbitMQ topic exchange, ALB) under peak concurrent load.

PROFESSIONAL EXPERIENCE

Stack Lane

Bangalore, India

Artificial Intelligence Intern

Mar 2025 – Jun 2025

- Fine-tuned LLaMA 2 in PyTorch with a custom training pipeline that **cut inference and training compute cost by 30%**.
- Refactored preprocessing scripts, increasing **data throughput by 35%** and enabling training on **datasets twice as large**.
- Led a 3-engineer team** building an AI chatbot from prototype to production delivery, owning architecture decisions and integration.

Vault of Codes

Remote

Web Development Intern

May 2024 – Jan 2025

- Built a responsive portfolio and e-commerce site with real-time chat and one-click checkout, improving cross-device **responsiveness by 30%** and **lifting conversion 20%**.
- Optimized page-load performance to **under 1.2 s** across devices via asset optimization and component-level lazy loading.

PUBLICATIONS

IEEE — Quantum-2Q Learning for DDoS Detection

2024

- Published an IEEE paper applying Quantum-2Q learning to DDoS detection across five IoT datasets, **cutting RL training time 35%** and reaching **95% attack-classification accuracy**.

TECHNICAL SKILLS

Languages: Go, Python, Java, C++, JavaScript, TypeScript, SQL

Concepts: Data Structures & Algorithms, Object-Oriented Design (OOP), Distributed Systems, Concurrency, CI/CD, Unit Testing, Agile/Scrum

Backend & Systems: FastAPI, Celery, RabbitMQ, REST APIs, WebSockets, Microservices, Docker, Kubernetes

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI & ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, sentence-transformers, LLM fine-tuning, LangChain

Cloud & DevOps: AWS (EC2, S3, Lambda), GitHub Actions, Linux, Prometheus, Grafana; React, Next.js, TailwindCSS